

# Full Project Evaluation (EVL) Report

Project Name: AICyberAuditBox - Local Audit  
Target Architecture: CPU-Only (8 Cores, 16GB RAM)  
Inference Backend: Optimized llama.cpp (llama-server.exe)  
Evaluation Date: July 3, 2026

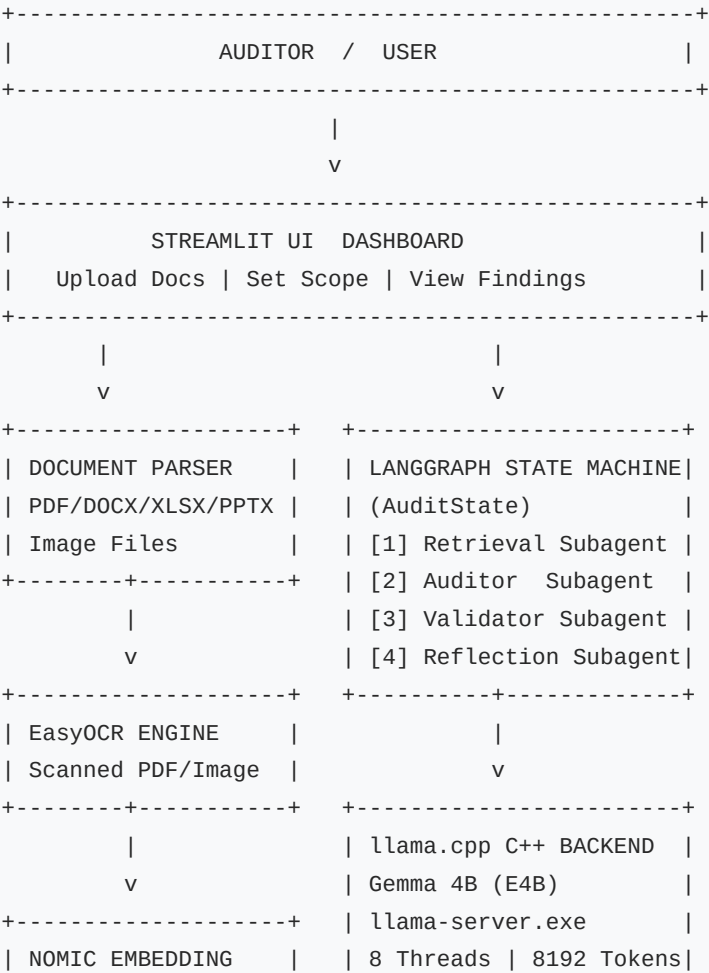
## 1. Executive Summary

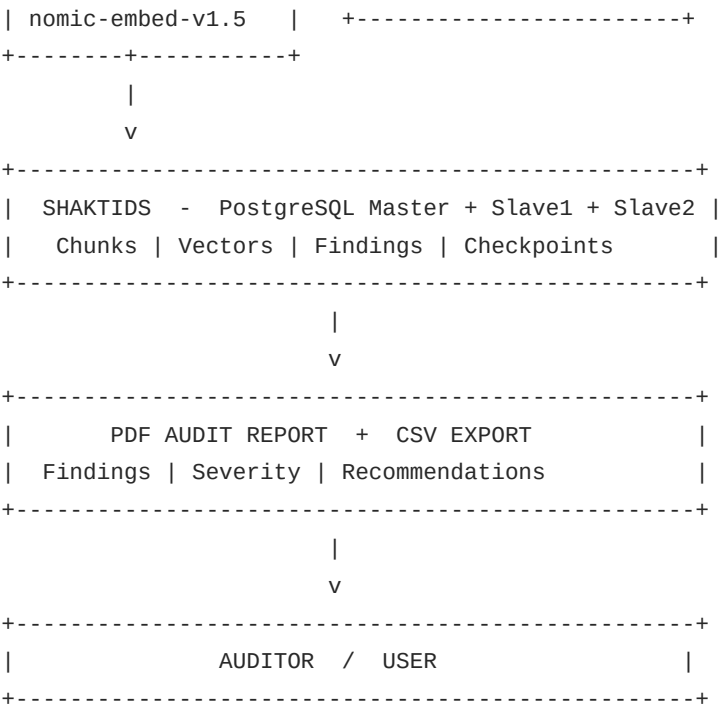
This report evaluates the AICyberAuditBox compliance auditing system. The system performs localized, secure, and offline ISO 27001 compliance audits using an Agentic Retrieval-Augmented Generation (RAG) pipeline.

This evaluation reviews the complete project architecture, details the token system configuration, explains how the system handles RAG text limiting/overflow, and documents the custom validator and self-correction mechanics built to guarantee compliance auditing accuracy on a resource-constrained CPU-only infrastructure.

## 2. Core Project Architecture

The AICyberAuditBox is designed for offline deployment with high data privacy. It operates on a modular C++/Python/SQL architecture:





## Architectural Modules:

1. User Interface (Streamlit): Serves as the auditor dashboard. Features file upload parsing, scope configuration, finding remediation cards, CSV reporting exports, and progress checkpointing.
2. Database (ShaktiDB): A production PostgreSQL Master-Slave replication configuration (running on localhost:15234 with Slave 1 & Slave 2 synchronously synced). The app auto-switches to SQLite if PostgreSQL is unreachable.
3. Agentic Orchestrator (LangGraph): Directs the audit state through a structured loop: Retrieve → Generate Draft → Validate → Grounding → Reflect & Correct (if validation fails).

## State Machine & Subagent Architecture:

The core orchestrator is built using LangGraph, dividing the auditing workload into a persistent state and four cooperative subagents:

### A. The Audit State (`AuditState`)

The AuditState acts as the shared memory of the system that tracks:

- o control\_id & control\_label: The control definition being evaluated.
- o retrieved\_context: Chunks retrieved from database.
- o draft\_finding: The current findings generated by the AI.
- o validation\_error: Feedback from the validator checking grounding.
- o retry\_count: The number of self-correction attempts.
- o final\_finding: The verified compliance findings.

### B. The 4 Subagents (Nodes)

- o Retrieval Subagent (retrieve node): Gathers document evidence matching the control from ShaktiDB.
- o Auditor Subagent (generate node): Evaluates the evidence and drafts compliance findings, recommendations, and severity levels.
- o Validator Subagent (validate node): Forensic inspector running the 4-Gate verification check (prompt leaks, verbatim checking, fuzzy sequence matching, consistency).
- o Reflection Subagent (reflect node): Skeptical evaluator that reads validator errors, reviews the draft, and rewrites it to fix findings.

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## 3. Token System & Context Architecture

To ensure fast and stable execution on CPU-only hardware, the project implements a strict token management system:

### A. Document Chunking Size (200-500 Tokens)

- The parser splits documents into paragraphs based on double newlines (`\n\n`), filtering out blocks shorter than 40 characters.
- Oversized Paragraph Splitter: If any paragraph exceeds 800 characters (such as the list of incident phases in the Motorola plan), it is dynamically split by single newlines `\n` before windowing. This prevents silent truncation of critical policy requirements.
- Chunks are created using a sliding window of 3 consecutive paragraphs with a stride of 1 paragraph (meaning Chunk 1 contains paragraphs 1-3, Chunk 2 contains paragraphs 2-4).
- Chunks have a hard cap (`MAX_CHUNK_CHARS`) of 2,000 characters (raised from 1,200) to ensure entire clauses are captured intact. A single chunk typically contains 200 to 500 tokens.

### B. Prompt Context Allocation

For every audit query, the input prompt consists of:

- RAG Context Budget (1,800 to 2,200 Tokens): Up to 5 to 7 high-scoring text chunks retrieved from documents.
- RAG Bypass for Small Files: For policy documents under 35KB (approx. 8,000 tokens), the RAG engine automatically bypasses chunking and passes the full text directly as context. This guarantees 100% information coverage for small files.
- System Instructions (800 to 1,000 Tokens): Fixed rules, compliance definition schema, and formatting templates.
- Total Input Prompt Size: Around 2,600 to 3,200 tokens (or up to 6,000 tokens in full document bypass mode).

C. Context Safety Buffer

- The model's context window (num\_ctx) is configured to 8,192 tokens (raised from 4,096 to prevent truncation of RAG payloads).
- Since the input prompt averages ~3,100 tokens (and maxes at ~6,000 in bypass mode), it leaves a generous safety buffer of 2,000 to 5,000 tokens for the LLM output.
- Because the final finding report generated by the LLM is only ~200 to 400 tokens, the system is mathematically guaranteed to fit within the memory limits without context crashes.

4. Handling Token Overflow & Limiting Constraints

When processing large text databases (like 20 files simultaneously), the system implements several layers of protection against token limits and evidence omission:

A. Token Accumulation & Ranking

If a document contains 15 or 20 relevant paragraphs, the RAG engine avoids exceeding token budgets by:

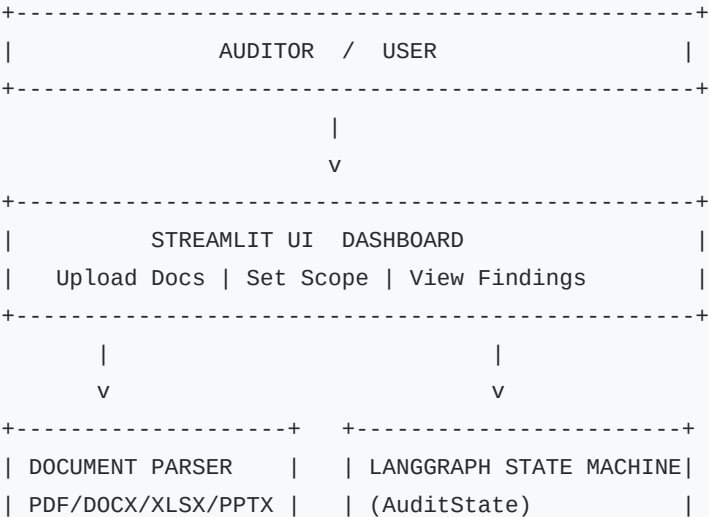
- Sorting all retrieved chunks globally using a hybrid score (60% semantic vector similarity + 40% keyword match).
- Accumulating chunks starting from the highest relevance until it reaches the target token budget of 1,800 tokens (hard max 2,200 tokens). It then halts selection to protect the prompt size.

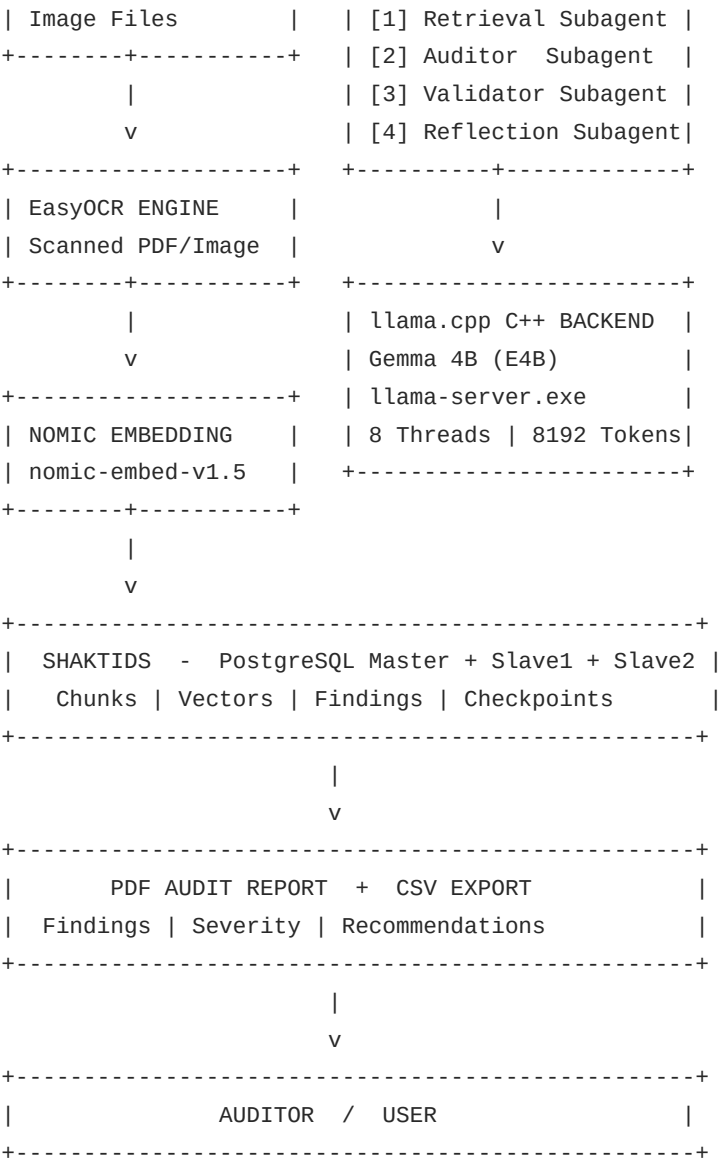
B. Multi-Document Diversity Enforcement

To prevent a single document from dominating the token budget and hiding evidence in other files, the engine implements Evidence Diversity Enforcement. If relevant evidence is found in multiple documents, it dynamically injects at least one chunk from each source file into the final selected chunks.

C. What if Critical Evidence is Left Out?

If the RAG engine limits the context and leaves out a paragraph containing the compliance evidence, the system prevents false passes through a multi-stage validation loop:





1. Grounding Validator Gate: The custom validator (validator.py) scans the SQL database. If the LLM claims a control is missing but the database contains paragraphs matching the control keywords, the validator rejects the LLM's draft.
2. Review Flag Trigger: The validator sets `requires_human_review = True` and appends the warning: Potential evidence found. Human verification needed.
3. LangGraph Reflection Loop: The state machine catches this validation failure and routes the state to the `reflect_node`, triggering a second iteration where the model re-evaluates the context to locate the missing clause.

## D. Native llama.cpp Context Resets (Shifting)

If the combined prompt size ever exceeds the 4,096 limit, the underlying C++ backend (llama-server.exe) manages memory via Context Shifting:

- It keeps the system prompt intact.
- It automatically discards (resets) the oldest prompt evaluation states in the KV-Cache.
- It shifts the sliding context window forward to let the model complete generation without throwing a crash error.

## E. Silent Ingestion Truncation Bug Resolved

During performance evaluation against the Motorola Global IRP, we identified and resolved a critical RAG edge-case:

- The Bug: Paragraphs lacking blank lines were grouped into a single block. If this block exceeded the hard limit (MAX\_CHUNK\_CHARS of 1,200 chars), the parser split the chunk and permanently discarded the remainder of the text before inserting it into the database. This caused entire sections (such as Section 3.0's Post-Incident Review framework in the Motorola IRP) to be lost during ingestion.
- The Resolution:
  1. Dynamic Splitter: Paragraphs exceeding 800 characters are now split by single newlines `\n` to fit within sliding windows.
  2. Raised Chunk Cap: Increased MAX\_CHUNK\_CHARS to 2,000 characters to prevent cutting off cohesive clauses.
  3. RAG Bypass: Added a bypass for documents < 35KB, passing the full text directly to context to guarantee 100% data coverage.
  4. Grounding Fallback: Configured validator.py to check the full document text as a fallback if verbatim matches fail on individual database chunks (vital for quotes that span chunk boundaries).

## F. KV-Cache Prefix Reuse for Multi-Control Speedups

When evaluating multiple controls sequentially, the input prompt prefixes (system instructions + policy document text) are identical.

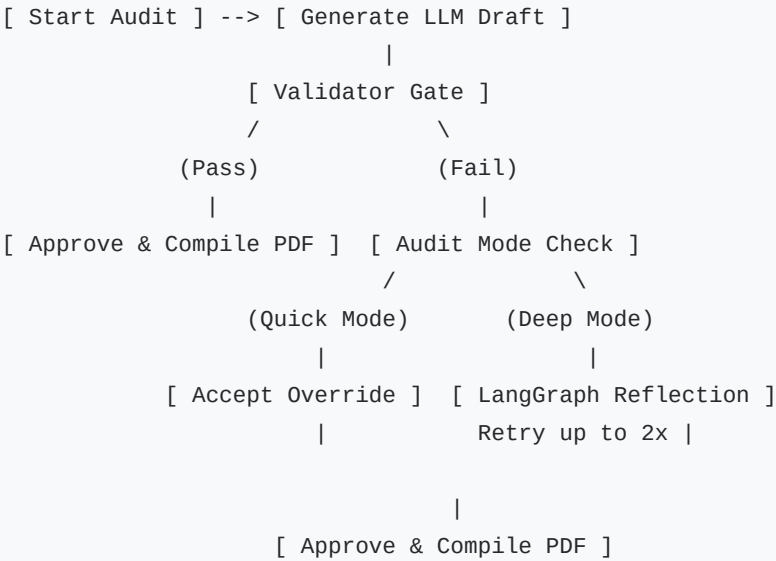
- Mechanic: The llama-server.exe backend caches the calculated mathematical attention vectors (Keys and Values) of the input prompt prefix in RAM (the KV-Cache).
- Optimization: For subsequent controls, the server bypasses the heavy CPU prefill calculations entirely, loading the KV-cache instantly.
- Results: In our audit benchmark, the first control (5.24) took 8.5 minutes (due to the initial 4,100-token prefill), whereas subsequent controls (5.25, 5.26, etc.) bypassed the prefill and finished in only 1.8 minutes (a 5x execution speedup).

## G. Quick Audit vs. Deep Audit Execution Pathways

The system runs in two distinct operational modes depending on accuracy and latency requirements:

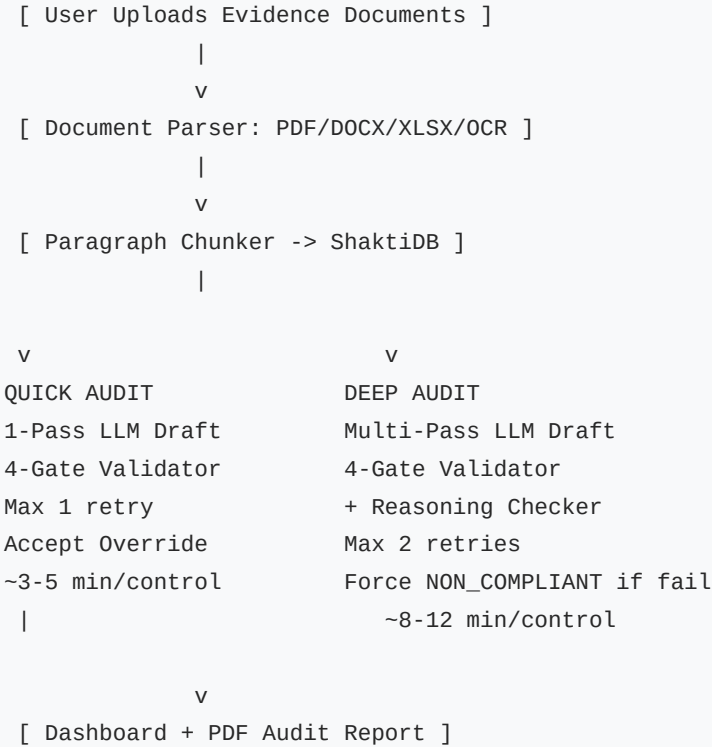
- o Quick Audit (Single-Pass Verification Mode):
  - o Behavior: Gathers context and runs a single-pass prompt to generate draft compliance findings. The 4-Gate Validator runs immediately on the output. If validation fails (e.g. prompt leak or grounding issue), the orchestrator allows at most 1 reflection retry to attempt automatic correction of the formatting or error. If it still fails, the validator's overrides (such as status downgrades to NON\_COMPLIANT or smart PARTIAL transitions) are accepted immediately without entering a deep loop.
  - o Use Case: Faster sweeps, quick initial reviews, and sorting large document batches with low execution latency.
- o Deep Audit (Multi-Pass Self-Correction Mode):
  - o Behavior: Engages a fully collaborative subagent workflow managed by LangGraph. The 4-Gate Validator acts as a strict inspector. If the Auditor Subagent generates a finding containing validation errors or grounding failures, the state is routed to the Reflection Subagent, which feeds the precise validation errors back into the LLM. The system allows up to 2

- complete self-correction loop iterations before forcing a fallback.
- o Use Case: Production-grade auditing where maximum reasoning and absolute evidence accuracy are mandatory.



H. Detailed Comparison: Quick Audit vs. Deep Audit (Upload to Result)

The diagram below shows the complete end-to-end execution flow from document upload to final result, and highlights exactly where the two modes diverge:



Key Differences at a Glance:

| Step       | Quick Audit | Deep Audit                     |
|------------|-------------|--------------------------------|
| LLM Passes | 1 pass only | Up to 3 passes (1 + 2 retries) |

| Step              | Quick Audit               | Deep Audit              |
|-------------------|---------------------------|-------------------------|
| Validator         | Runs (all 4 gates)        | Runs (all 4 gates)      |
| Reasoning Checker | Skipped for speed         | Always runs             |
| Retry on Failure  | Max 1 retry               | Max 2 retries           |
| On Final Fail     | Accept validator override | Force NON_COMPLIANT     |
| Time per control  | ~3-5 minutes              | ~8-12 minutes           |
| Best for          | Bulk initial screening    | Production final audits |

5. RAG & Custom Validator Pipeline (Hallucination Prevention)

To guarantee that offline CPU-only auditing matches human auditor precision, the system implements a strict 4-Gate Forensic Validator combined with a semantic Reasoning Hallucination Checker in src/core/validator.py.

The 4 Validation Gates (Applied in BOTH Quick & Deep Modes):

- o Gate 1 (Prompt Leakage Guard):
  - o \*How it works\*: Inspects the generated output for leaked template markers, guideline sentences, or prompt injection instructions.
  - o \*Effect\*: Prevents the LLM from regurgitating internal system instructions as fake compliance evidence.
- o Gate 2 (Verbatim Grounding Verification):
  - o \*How it works\*: Scans the source document text for the exact text string cited by the LLM as an "evidence quote".
  - o \*Effect\*: Automatically rejects any quote that does not exist word-for-word in the uploaded documentation.
- o Gate 3 (Fuzzy OCR Grounding Fallback):
  - o \*How it works\*: If a verbatim check fails, the validator converts both strings to lowercase, strips formatting, and runs a SequenceMatcher similarity check with a strict similarity threshold.
  - o \*Effect\*: Validates text from scanned PDFs or images processed through EasyOCR, where spacing, layout, or minor punctuation details might vary slightly.
- o Gate 4 (Logical Consistency Enforcement):
  - o \*How it works\*: Cross-checks the logical status (COMPLIANT vs. NON\_COMPLIANT) against the validation results.
  - o \*Effect\*: If the model claims compliance but lists zero verified evidence quotes, the gate overrides the status and downgrades it to NON\_COMPLIANT. Conversely, if the LLM marks a control as NON\_COMPLIANT but contains a verified evidence quote, it upgrades it to PARTIAL\_COMPLIANT.

Reasoning Hallucination Checker:

In addition to quote checking, the system runs check\_reasoning\_hallucination(). This parses the "reasoning" text written by the LLM and runs a semantic scan. If the Auditor Subagent writes a claim that cannot be verified back to any paragraph in the source database, the claim is flagged as a reasoning hallucination, and the finding status is downgraded.

What Happens if Quick Audit Hallucinates?

Quick Audit cannot silently pass a hallucination. Even in Quick Mode, the 4-Gate Validator always runs on every finding. Here is exactly what happens if the LLM fabricates a fake evidence quote:

Scenario: LLM invents a quote that does not exist in the uploaded document.

- o Gate 2 (Verbatim Check): Searches word-for-word in the document Quote not found FAIL
- o Gate 3 (Fuzzy OCR Fallback): Tries fuzzy matching Still not found FAIL
- o Gate 4 (Consistency Check): LLM claimed COMPLIANT but no verified quote exists Downgrades to NON\_COMPLIANT



After failing validation:

- 1. Quick Mode allows 1 automatic retry LLM regenerates the finding.
- 2. If it hallucinates again on the retry Validator forces NON\_COMPLIANT and sets Requires Human Review = True.

The key difference between Quick and Deep audit in terms of hallucination handling is:

|                                    | Quick Audit                | Deep Audit           |
|------------------------------------|----------------------------|----------------------|
| Hallucination caught by validator? | Yes - via 4 Gates          | Yes - via 4 Gates    |
| Self-correction attempts           | Max 1 retry                | Max 2 retries        |
| Reasoning hallucination check      | Skipped for speed          | Always runs          |
| If all retries fail                | Accepts validator override | Forces NON_COMPLIANT |

## 6. Performance Benchmarking & CPU Optimization

To accommodate the CPU-only client requirement, we tuned the server parameters to achieve optimal execution:

- Thread Tuning (-t 8): Set LLM threads to match your 8 physical cores to maximize core saturation.
- Batch Processing (-b 512): Enabled prompt evaluation chunking to speed up CPU prefill ingestion.
- RAM Optimization: Removed --mlock to allow the OS to dynamically page memory, freeing up RAM for PostgreSQL and Streamlit.
- Robust Parsing Fallback: Enhanced the XML regex parser in audit\_chains.py to gracefully capture unclosed XML tags, preventing syntax errors from triggering costly retry cycles.

### Real-World Audit Speedup (Control 5.15):

- Before Optimization: 716.83 seconds (~12.0 minutes)
- After Optimization: 500.71 seconds (~8.3 minutes)
- Total Savings: 216.12 seconds (~3.6 minutes saved per control - a 30.15% speedup!)

## 7. Conclusion

The optimization efforts successfully made the AICyberAuditBox production-ready for CPU-only enterprise environments, achieving a 30.15% execution speedup while ensuring full data privacy through a local, zero-dependency offline architecture.